

Manage your Infrastructure using code

Uma breve introdução a AWS e Terraform

Olá!



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🧐 Nerd

♟️ Jogador de Xadrez

🏆 Desportista sub-amador

Agenda

1. O que é computação em Cloud?
2. Introdução à AWS — **A**ma**z**on **W**eb **S**ervices
3. Problemas com gerir infraestrutura na AWS
4. Entra IaC — Porquê usar código?
5. Demonstração de Terraform
6. Alternativas
7. Recursos adicionais

0 que é computação em Cloud?

A computação em Cloud é a disponibilização de recursos computacionais através da internet:

- Servidores
- Armazenamento
- Redes
- Bases de dados
- Software

0 que é computação em Cloud?

Traz vantagens como:

- Permitir o acesso democratizado através de qualquer parte do mundo
- Remover a necessidade de adquirir recursos físicos
- Escalabilidade dos recursos conforme necessário
- Flexibilidade e eficiência de custos
- Inovação — Facilidade de exploração!

0 que é computação em Cloud?



Introdução à AWS — Amazon Web Services

- Surgiu em 2006
- Mais de 200 tipos de serviços disponíveis
- Data centers em mais de 25 regiões segregadas no mundo todo
- Detentores de cerca de 30% a 35% do mercado de computação Cloud
- Usada por gigantes como Airbnb, Netflix, GoDaddy, Starbucks, Toyota e Kevel

Introdução à AWS — Amazon Web Services

Exemplos de soluções Cloud disponibilizadas:

- Servidores — EC2, ECS
- Armazenamento — S3, EBS
- Redes — VPC, CloudFront
- Bases de dados — RDS, DynamoDB
- Software — Lambda, SQS

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Introdução à AWS — Amazon Web Services

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Introdução à AWS — Amazon Web Services

Exemplos de soluções Cloud disponibilizadas:

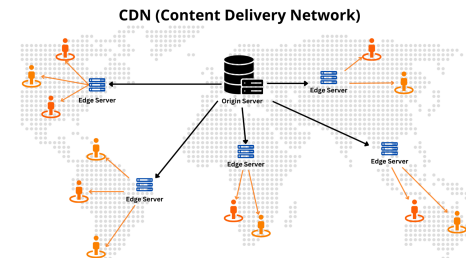
- Servidores — EC2, ECS
- Armazenamento — S3, EBS
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- Bases de dados — RDS, DynamoDB
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- Software — Lambda, SQS



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- Bases de dados — RDS, **DynamoDB**
- Software — Lambda, SQS



Introdução à AWS — Amazon Web Services

Exemplos de soluções Cloud disponibilizadas:

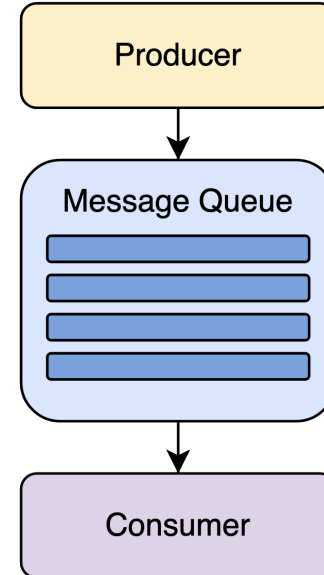
- Servidores — EC2, ECS
- Armazenamento — S3, EBS
- Redes — VPC, CloudFront
- Bases de dados — RDS, DynamoDB
- Software — **Lambda**, SQS



Introdução à AWS — Amazon Web Services

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- Redes — VPC, CloudFront
- Bases de dados — RDS, DynamoDB
- Software — Lambda, **SQS**



Introdução à AWS — Máquina virtual

☰

EC2

<

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EC2 Global View

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▼ Instances

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▼ Images

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▼ Elastic Block Store

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Lifecycle Manager

▼ Network & Security

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Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Resources

EC2 Global View

⚙️

🔄

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	0	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	0
Key pairs	3	Load balancers	0	Placement groups	0
Security groups	5	Snapshots	0	Volumes	0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Instance alarms

View in CloudWatch

⚠️ 0 in alarm

✅ 0 OK

⌚ 0 insufficient data

Instances in alarm

Scheduled events

🔄

United States (N. Virginia)

No scheduled events

Service health

AWS Health Dashboard

🔄

Region

United States (N. Virginia)

Status

✅ This service is operating normally.

Zones

Zone name	Zone ID
us-east-1a	use1-az4
us-east-1b	use1-az6
us-east-1c	use1-az1
us-east-1d	use1-az2
us-east-1e	use1-az3
us-east-1f	use1-az5

Enable additional Zones

Account attributes

🔄

Default VPC

vpc-0527b79bae110e37d

Settings

Data protection and security

Allowed AMIs

Zones

EC2 Serial Console

Default credit specification

EC2 console preferences

Explore AWS

✕

Get Up to 40% Better Price Performance

T4g instances deliver the best price performance for burstable general purpose workloads in Amazon EC2.

Learn more

Amazon GuardDuty Malware Protection

GuardDuty now provides agentless malware detection in Amazon EC2 & EC2 container workloads.

Learn more

10 Things You Can Do Today to Reduce AWS Costs

Explore how to effectively manage your AWS costs without compromising on performance or capacity.

Learn more

Additional information

🔗

Get started walkthroughs

Getting started guide

Documentation

All EC2 resources

Introdução à AWS — Máquina virtual

The screenshot displays the AWS Management Console interface for the EC2 Instances page. On the left, a navigation sidebar lists various AWS services and EC2-specific options. The main area shows the 'Instances' page with a search bar, filters, and a table of instances. The filter 'Instance state = running' is applied, and the table indicates 'No matching instances found'.

EC2 > Instances

Instances [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

[Instance state = running](#) [Clear filters](#) < 1 > [Settings](#)

☐	Name 🔗	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
No matching instances found								

Select an instance [Settings](#) [Dropdown](#)

Navigation Sidebar:

- EC2
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Introdução à AWS — Máquina virtual

EC2 > Instances > Launch an instance



Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

enei-prsentation-service

[Add additional tags](#)

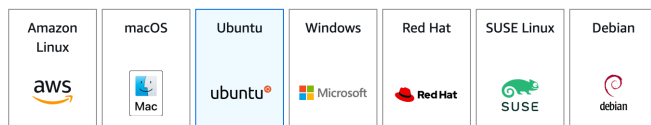
▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

My AMIs

Quick Start



[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-084568db4383264d4 (64-bit (x86)) / ami-0c4e709339fa8521a (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

64-bit (x86)

AMI ID

ami-084568db4383264d4

Publish Date

2025-03-05

Username [Info](#)

ubuntu

Verified provider

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-084568db4383264d4

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Launch instance

[Preview code](#)

Introdução à AWS — Máquina virtual

EC2 > Instances > Launch an instance



Success

Successfully initiated launch of instance (i-034e838801daea7bf)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

< 1 2 3 4 5 6 >

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

Introdução à AWS — Máquina virtual

☰

EC2 > Instances > i-034e838801daea7bf

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📄

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Instance summary for i-034e838801daea7bf (enei-prsentation-service) [Info](#)

Updated less than a minute ago

Instance ID

[i-034e838801daea7bf](#)

IPv6 address

—

Hostname type

IP name: ip-172-31-88-164.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

[35.170.202.79](#) [Public IP]

IAM Role

[jf-allow-ssm-sessions-access](#)

IMDSv2

Required

Operator

—

Public IPv4 address

[35.170.202.79](#) | [open address](#)

Instance state

🟢 Running

Private DNS name (IPv4 only)

[ip-172-31-88-164.ec2.internal](#)

Instance type

t2.micro

VPC ID

[vpc-0527b79bae110e37d](#)

Subnet ID

[subnet-0714b7ecdbab05666](#)

Instance ARN

[arn:aws:ec2:us-east-1:891377335563:instance/i-034e838801daea7bf](#)

Private IPv4 addresses

[172.31.88.164](#)

Public IPv4 DNS

[ec2-35-170-202-79.compute-1.amazonaws.com](#) | [open address](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

No recommendations available for this instance.

Auto Scaling Group name

—

Managed

false

🔄 Connect

Instance state ▼

Actions ▼

Details

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▼ Instance details [Info](#)

AMI ID

[ami-084568db4383264d4](#)

AMI name

[ubuntu/images/hvm-ssd-gp3/ubuntu-noble-24.04-amd64-server-20250305](#)

Stop protection

Disabled

Monitoring

disabled

Allowed image

—

Launch time

[Sat Mar 22 2025 10:10:10 GMT+0000 \(Western European](#)

Platform details

[Linux/UNIX](#)

Termination protection

Disabled

AMI location

[amazon/ubuntu/images/hvm-ssd-gp3/ubuntu-noble-2](#)

Introdução à AWS — Máquina virtual

```
root@ip-172-31-88-164:/home/ubuntu# ls -la
total 24
drwxr-x--- 3 ubuntu ubuntu 4096 Mar 22 10:10 .
drwxr-xr-x 4 root    root   4096 Mar 22 10:11 ..
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx----- 2 ubuntu ubuntu 4096 Mar 22 10:10 .ssh
root@ip-172-31-88-164:/home/ubuntu#
```

Introdução à AWS — Máquina virtual

```
root@ip-172-31-88-164:/home/ubuntu# snap install docker
docker 27.5.1 from Canonical✓ installed
root@ip-172-31-88-164:/home/ubuntu# docker pull postgres:alpine
alpine: Pulling from library/postgres
f18232174bc9: Pull complete
d8d8fb695a5a: Pull complete
c24c1ba610df: Pull complete
83efd74bc97e: Pull complete
215ba3ecdc26: Pull complete
15ca4c67ed92: Pull complete
4f18957d9158: Pull complete
404c53e09e31: Pull complete
9e4acb9ca7d3: Pull complete
8f4971a5dfe7: Pull complete
Digest: sha256:7062a2109c4b51f3c792c7ea01e83ed12ef9a980886e3b3d380a7d2e5f6ce3f5
Status: Downloaded newer image for postgres:alpine
docker.io/library/postgres:alpine
root@ip-172-31-88-164:/home/ubuntu# touch env.txt
root@ip-172-31-88-164:/home/ubuntu# echo 'PORT=8080' >> env.txt
root@ip-172-31-88-164:/home/ubuntu# echo 'ENV=test' >> env.txt
root@ip-172-31-88-164:/home/ubuntu#
```

Introdução à AWS — Máquina virtual

[EC2](#) > [Security Groups](#) > Create security group



Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

VPC [Info](#)

Inbound rules [Info](#)

Type [Info](#)

Protocol [Info](#)

Port range [Info](#)

Source [Info](#)

Description - optional [Info](#)

Allow HTTPS traffic from the internet

Introdução à AWS — Máquina virtual

☰

EC2 > Security Groups > sg-012c22aac5890d813 - enei-security-group

🔄 📄

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Target Groups

sg-012c22aac5890d813 - enei-security-group

Actions ▼

Details

Security group name

❏ enei-security-group

Security group ID

❏ sg-012c22aac5890d813

Description

❏ enei-security-group

VPC ID

❏ vpc-0527b79bae110e37d [🔗](#)

Owner

❏ 891377335563

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

🔄 Manage tags

Edit inbound rules

🔍 Search

< 1 > ⚙️

☐	Name ▼	Security group rule ID ▼	IP version ▼	Type ▼	Protocol ▼	Port range ▼
☐	-	sgr-0a2771ab462093226	IPv4	HTTPS	TCP	443

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EC2

Instances

i-034e838801daea7bf

Instance summary for i-034e838801daea7bf (enei-prsentation-service)

Updated 12 minutes ago

Instance ID

i-034e838801daea7bf

IPv6 address

–

Hostname type

IP name: ip-172-31-88-164.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

35.170.202.79 [Public IP]

IAM Role

jf-allow-ssm-sessions-access

IMDSv2

Required

Operator

–

Public IPv4 address

35.170.202.79 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-88-164.ec2.internal

Instance type

t2.micro

VPC ID

vpc-0527b79b9ae110e37d

Subnet ID

subnet-0714b7ecdab05666

Instance ARN

arn:aws:ec2:us-east-1:891377335563:instance/i-034e838801daea7bf

Connect

Instance state

Actions

Connect

Manage instance state

Instance settings

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Security

Image and templates

Monitor and troubleshoot

Private IPv4 addresses

172.31.88.164

Public IPv4 DNS

ec2-35-170-202-79-54-172-31-88-164.ec2.internal

Change security groups

Get Windows password

Modify IAM role

Elastic IP addresses

–

AWS Compute Optimizer finding

No recommendations available for this instance.

Auto Scaling Group name

–

Managed

false

Details

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Security details

IAM Role

jf-allow-ssm-sessions-access

Owner ID

891377335563

Launch time

Sat Mar 22 2025 10:10:10 GMT+0000 (Western European Standard Time)

Security groups

sg-08a9e6e65586f18ec (launch-wizard-5)

Introdução à AWS — Máquina virtual

EC2 > Instances > i-034e838801daea7bf > Change security groups



Change security groups [Info](#)

Amazon EC2 evaluates all the rules of the selected security groups to control inbound and outbound traffic to and from your instance. You can use this window to add and remove security groups.

Instance details

Instance ID

i-034e838801daea7bf

Network interface ID

eni-052b88ed303024fca

Associated security groups

Add one or more security groups to the network interface. You can also remove security groups.

Q sg-012c22aac5890d813



Add security group

Security groups associated with the network interface (eni-052b88ed303024fca)

Security group ID	Security group name	Description	Owner ID	
sg-012c22aac5890d813	enei-security-group	enei-security-group	891377335563	Remove

[Cancel](#)

[Save](#)

Introdução à AWS — Máquina virtual



Alguns problemas!

- Processo moroso — minutos!
- Difícil "apanhar" modificações aos recursos
- Incapacidade de replicabilidade
- Falta de versionamento dos recursos
- Difícil trabalhar colaborativamente
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?

Alguns problemas?

- Processo moroso — minutos!
- Difícil "apanhar" modificações aos recursos
- Incapacidade de replicabilidade
- **Falta de versionamento dos recursos**
- **Difícil trabalhar colaborativamente**
- **Falta de documentação escrita — Porquê que criamos cada um dos recursos?**

Desenvolvimento de código — Alguns problemas?

- Falta de versionamento dos recursos
- Difícil trabalhar colaborativamente
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?

Estes problemas são resolvidos por ferramentas de versionamento de código como **git**!

Se conseguíssemos descrever a nossa **infraestrutura**
utilizando código, teríamos alguns destes problemas
automaticamente resolvidos!



Terraform

0 que é Terraform?

- Ferramenta de definição de infraestrutura utilizando código (Infrastructure as Code — **IaC**)
- Permite definir recursos (máquinas virtuais, bases de dados) declarativamente
 - Definimos **o que queremos**, não como o que queremos é criado
- Surgiu em **2014**, criado pela HashiCorp
- Suporta muitos **providers** diferentes, tal como:
 - Azure, GCP, **AWS**
 - Kubernetes, Helm, StatusCake, GitHub, Datadog

Exemplo

Vamos criar exatamente os mesmos recursos de há pouco, mas utilizando código declarativo **Terraform** para os definir!

1. Criar máquina virtual (AWS **EC2**)
2. Adicionar comandos de instalação e configuração de ferramentas dentro da máquina virtual (docker, postgres, ...)
3. Criar regra de firewall que permita entrada tráfego HTTPS na nossa máquina virtual (AWS **SecurityGroup**)
4. Associar a regra de firewall à nossa máquina virtual

Configuração

```
# config.tf

terraform {
  required_version = "1.8.7"

  required_providers {
    aws = {
      source = "hashicorp/aws"
    }
  }
}

provider "aws" {
  region = "us-east-1"
}
```

Inicialização

```
> terraform init
```

```
Initializing the backend...
```

```
Initializing provider plugins...
```

- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v5.92.0...
- Installed hashicorp/aws v5.92.0 (signed by HashiCorp)

```
...
```

```
Terraform has been successfully initialized!
```

Inicialização

```
> ls -la
```

```
drwxr-xr-x  3 ralves  staff    96 22 Mar 11:33 .terraform
-rw-r--r--  1 ralves  staff 1377 22 Mar 11:33 .terraform.lock.hcl
-rw-r--r--  1 ralves  staff  168 22 Mar 11:33 config.tf
```

Inicialização

```
> tree .terraform
.terraform
├── providers
│   ├── registry.terraform.io
│   │   ├── hashicorp
│   │   │   ├── aws
│   │   │   │   ├── 5.92.0
│   │   │   │   │   ├── darwin_arm64
│   │   │   │   │   │   ├── LICENSE.txt
│   │   │   │   │   │   └── terraform-provider-aws_v5.92.0_x5
```

Inicialização

```
> cat .terraform.lock.hcl
# This file is maintained automatically by "terraform init".
# Manual edits may be lost in future updates.

provider "registry.terraform.io/hashicorp/aws" {
  version = "5.92.0"
  hashes = [
    "h1:Hm5w8euRSm6tZyc60+nVPQheCikB7P0NhFI/dSFK0IM=",
    "zh:1d3a0b40831360e8e988aee74a9ff3d69d95cb541c2eae5cb843c64303a091ba",
    "zh:3d29cbced6c708be2041a708d25c7c0fc22d09e4d0b174360ed113bfae786137",
    "zh:4341a203cf5820a0ca18bb514ae10a6c113bc6a728fb432acbf817d232e8eff4",
    "zh:54c7189358b37fd895dedbabf84e509c1980a8c404a1ee5b29b06e40497b8655",
    "zh:5d8bb1ff089c37cb65c83b4647f1981fded993e87d8132915d92d79f29e2fcd8",
    ...
  ]
}
```

1. Criar máquina virtual

```
# ec2.tf

resource "aws_instance" "virtual-machine" {
  ami           = "ami-0c7217cdde317cfec" # Ubuntu Server 22.04
  instance_type = "t2.micro"

  tags = {
    Name = "enei-presentation-instance"
  }
}
```

1. Criar máquina virtual — Planear

```
> terraform plan
```

Terraform will perform **the** following actions:

```
# aws_instance.virtual-machine will be created
+ resource "aws_instance" "virtual-machine" {
  + ami                    = "ami-0c7217cdde317cfec"
  + instance_type         = "t2.micro"
  + tags                  = {
    + "Name" = "enei-presentation-instance"
  }
  + id                   = (known after apply)
  + arn                  = (known after apply)
  ...
}
```

Plan: **1** to add, **0** to change, **0** to destroy.

1. Criar máquina virtual — Aplicar

```
> terraform apply
```

```
Terraform will perform the following actions:
```

```
...
```

```
Plan: 1 to add, 0 to change, 0 to destroy.
```

```
Do you want to perform these actions?
```

```
  Terraform will perform the actions described above.
```

```
  Only 'yes' will be accepted to approve.
```

```
Enter a value:
```

1. Criar máquina virtual — Aplicar

```
> terraform apply
```

```
Terraform will perform the following actions:
```

```
...
```

```
Plan: 1 to add, 0 to change, 0 to destroy.
```

```
Do you want to perform these actions?
```

```
  Terraform will perform the actions described above.
```

```
  Only 'yes' will be accepted to approve.
```

```
Enter a value: yes
```

```
aws_instance.virtual-machine: Creating...
```

```
aws_instance.virtual-machine: Still creating... [10s elapsed]
```

```
aws_instance.virtual-machine: Creation complete after 14s [id=i-0c155532ff05b]
```

```
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

Estado

Terraform mantém um **estado** interno:

- Faz um mapeamento entre o "mundo real" e o código escrito
- A criação, remoção ou alteração de um recurso resulta numa mudança no estado
- Permite calcular diferenças entre os recursos existentes e o código existente
- Pode ser guardado **localmente** ou remotamente

Estado

```
> cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.10.3",
  "resources": [{
    "type": "aws_instance",
    "name": "virtual-machine",
    "provider": "provider[\"registry.terraform.io/hashicorp/aws\"]",
    "attributes": {
      "arn": "arn:aws:ec2:us-east-1:891377335563:instance/i-0c155532ff05b4f6",
      "id": "i-0c155532ff05b4f6",
      "tags": {
        "Name": "enei-presentation-instance"
      }
    }
  ]
}
```

Estado - Novo plano

```
> terraform plan
```

```
aws_instance.virtual-machine: Refreshing state... [id=i-0c155532ff05b4f6]
```

```
No changes. Your infrastructure matches the configuration.
```

```
Terraform has compared your real infrastructure against your configuration and  
found no differences, so no changes are needed.
```

2. Adicionar comandos — script

```
# user-data.sh

#!/bin/bash

# Set up docker
snap install docker

# Retrieve postgres docker image
docker pull postgres:alpine

# Configure env file
touch env.txt
echo 'PORT=8080' >> env.txt
echo 'ENV=test' >> env.txt
```

2. Adicionar comandos — Mudar código

```
# ec2.tf

resource "aws_instance" "virtual-machine" {
  ami          = "ami-0c7217cdde317cfec" # Ubuntu Server 22.04
  instance_type = "t2.micro"

  tags = {
    Name = "enei-presentation-instance"
  }
}
```

2. Adicionar comandos — Mudar código

```
# ec2.tf

resource "aws_instance" "virtual-machine" {
  ami           = "ami-0c7217cdde317cfec" # Ubuntu Server 22.04
  instance_type = "t2.micro"

  user_data = file("user-data.sh")

  tags = {
    Name = "enei-presentation-instance"
  }
}
```

2. Adicionar comandos — Planear

```
> terraform plan
```

```
aws_instance.virtual-machine: Refreshing state... [id=i-0c155532ff05b4f6]
```

Terraform will perform the following actions:

```
# aws_instance.virtual-machine will be updated in-place
```

```
~ resource "aws_instance" "virtual-machine" {
```

```
  + user_data                               = "10f6c789eb22e44d2420bc5bb..."
```

```
  # (37 unchanged attributes hidden)
```

```
}
```

```
Plan: 0 to add, 1 to change, 0 to destroy.
```

2. Adicionar comandos — Aplicar

```
> terraform apply
aws_instance.virtual-machine: Refreshing state... [id=i-0c155532ff05b4f6]

Terraform will perform the following actions:

...

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.virtual-machine: Modifying... [id=i-0c155532ff05b4f6]
aws_instance.virtual-machine: Still modifying... [id=i-0c155532ff05b4f6]
aws_instance.virtual-machine: Modifications complete after 12s

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.
```

3. Criar SecurityGroup — Código

```
# security-group.tf

resource "aws_security_group" "enei-security-group" {
  name          = "enei-security-group"
  description    = "Allow HTTPS inbound traffic"

  ingress {
    description = "HTTPS"
    from_port   = 443
    to_port     = 443
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"] # Allow from anywhere
  }
}
```

3. Criar SecurityGroup — Planear e aplicar

```
> terraform apply
aws_instance.virtual-machine: Refreshing state... [id=i-0c155532ff05b4f6]
```

Terraform will perform the following actions:

```
# aws_security_group.enei-security-group will be created
+ resource "aws_security_group" "enei-security-group" {
  + name = "enei-security-group"
  ...
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Enter a value: yes

```
aws_security_group.enei-security-group: Creating...
```

```
aws_security_group.enei-security-group: Creation complete after 4s
```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

4. Associar SG — Código

```
# ec2.tf

resource "aws_instance" "virtual-machine" {
  ami           = "ami-0c7217cdde317cfec" # Ubuntu Server 22.04
  instance_type = "t2.micro"

  user_data = file("user-data.sh")

  tags = {
    Name = "enei-presentation-instance"
  }
}
```

4. Associar SG — Código

```
# ec2.tf

resource "aws_instance" "virtual-machine" {
  ami           = "ami-0c7217cdde317cfec" # Ubuntu Server 22.04
  instance_type = "t2.micro"

  user_data = file("user-data.sh")

  vpc_security_group_ids = [
    aws_security_group.enei-security-group.id
  ]

  tags = {
    Name = "enei-presentation-instance"
  }
}
```

4. Associar SG — Planear e aplicar

```
> terraform apply
```

```
aws_security_group.enei-security-group: Refreshing state... [id=sg-0f3709f1bc466
```

```
aws_instance.virtual-machine: Refreshing state... [id=i-0c1555532ff05b4f6]
```

Terraform will perform the following actions:

```
# aws_instance.virtual-machine will be updated in-place
```

```
~ resource "aws_instance" "virtual-machine" {
```

```
  + vpc_security_group_ids = [
```

```
    + "sg-0f3709f1bc4662aef"
```

```
  ]
```

```
  # (39 unchanged attributes hidden)
```

```
}
```

Plan: 0 to add, 1 to change, 0 to destroy.

...

4. Associar SG — Planear e aplicar

...

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_instance.virtual-machine: Modifying... [id=i-0c155532ff05b4f6]

aws_instance.virtual-machine: Modifications complete after 3s [id=i-0c155532ff05b4f6]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Sucesso!?



Problemas solucionados?

- Incapacidade de replicabilidade
- Falta de versionamento dos recursos
- Difícil trabalhar colaborativamente
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?
- Processo moroso — minutos!
- Difícil "apanhar" modificações aos recursos

Problemas solucionados?

- **Incapacidade de replicabilidade**
- Falta de versionamento dos recursos
- Difícil trabalhar colaborativamente
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?
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Problemas solucionados?

- Incapacidade de replicabilidade 
- **Falta de versionamento dos recursos**
- Difícil trabalhar colaborativamente
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?
- Processo moroso — minutos!
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Problemas solucionados?

- Incapacidade de replicabilidade ✓
- Falta de versionamento dos recursos ✓
- **Difícil trabalhar colaborativamente**
- Falta de documentação escrita — Porquê que criamos cada um dos recursos?
- Processo moroso — minutos!
- Difícil "apanhar" modificações aos recursos

Problemas solucionados?

- Incapacidade de replicabilidade ✓
- Falta de versionamento dos recursos ✓
- Difícil trabalhar colaborativamente ✓
- **Falta de documentação escrita — Porquê que criamos cada um dos recursos?**
- Processo moroso — minutos!
- Difícil "apanhar" modificações aos recursos

Problemas solucionados?

- Incapacidade de replicabilidade 
- Falta de versionamento dos recursos 
- Difícil trabalhar colaborativamente 
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- **Processo moroso — minutos!**
- Difícil "apanhar" modificações aos recursos

Processo moroso

Há uma curva de aprendizagem.

Mas existe boa documentação **detalhada** para a maior parte dos providers!

Documentação – Exemplo **aws_instance**

 Registry [Browse](#) [Publish](#) [Sign-in](#)  [Use HCP Terraform for free](#)

Providers / [hashicorp](#) / [aws](#) / Version 5.92.0 [Latest Version](#)

aws  [Overview](#) [Documentation](#) [USE PROVIDER](#)

AWS DOCUMENTATION

- aws provider
 - Guides
 - Functions
 - ACM (Certificate Manager)
 - ACM PCA (Certificate Manager Private Certificate Authority)
 - AMP (Managed Prometheus)
 - API Gateway
 - API Gateway V2
 - Account Management
 - Amazon Q Business
 - Amplify
 - App Mesh
 - App Runner
 - AppConfig
 - AppFabric
 - AppFlow
 - AppIntegrations
 - AppStream 2.0

Resource: aws_instance

Provides an EC2 instance resource. This allows instances to be created, updated, and deleted. Instances also support [provisioning](#).

Example Usage

Basic example using AMI lookup

```
data "aws_ami" "ubuntu" {
  most_recent = true

  filter {
    name   = "name"
    values = ["ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-*"]
  }

  filter {
    name   = "virtualization-type"
    values = ["hvm"]
  }

  owners = ["099720109477"] # Canonical
}

resource "aws_instance" "web" {
  ami           = data.aws_ami.ubuntu.id
  instance_type = "t3.micro"
}
```

[Copy](#)

New Multi-language provider docs

Terraform

The Registry now supports multi-language docs powered by CDK for Terraform. [Learn more](#)

ON THIS PAGE

- [Example Usage](#)
- Tag Guide
- Argument Reference
- Attribute Reference
- Timeouts
- Import

[Report an issue](#)

Documentação – Exemplo `aws_instance`

AWS DOCUMENTATION

Q Filter

aws provider

- > Guides
- > Functions
- > ACM (Certificate Manager)
- > ACM PCA (Certificate Manager Private Certificate Authority)
- > AMP (Managed Prometheus)
- > API Gateway
- > API Gateway V2
- > Account Management
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- > AppConfig
- > AppFabric
- > AppFlow
- > AppIntegrations
- > AppStream 2.0
- > AppSync
- > Application Auto Scaling
- > Athena
- > Audit Manager
- > Auto Scaling
- > Auto Scaling Plans

Argument Reference

This resource supports the following arguments:

- `ami` - (Optional) AMI to use for the instance. Required unless `launch_template` is specified and the Launch Template specifies an AMI. If an AMI is specified in the Launch Template, setting `ami` will override the AMI specified in the Launch Template.
- `associate_public_ip_address` - (Optional) Whether to associate a public IP address with an instance in a VPC.
- `availability_zone` - (Optional) AZ to start the instance in.
- `capacity_reservation_specification` - (Optional) Describes an instance's Capacity Reservation targeting option. See [Capacity Reservation Specification](#) below for more details.

NOTE:

Changing `cpu_core_count` and/or `cpu_threads_per_core` will cause the resource to be destroyed and re-created.

- `cpu_core_count` - (Optional, **Deprecated** use the `cpu_options` argument instead) Sets the number of CPU cores for an instance. This option is only supported on creation of instance type that support CPU Options [CPU Cores and Threads Per CPU Core Per Instance Type](#) - specifying this option for unsupported instance types will return an error from the EC2 API.
- `cpu_options` - (Optional) The CPU options for the instance. See [CPU Options](#) below for more details.
- `cpu_threads_per_core` - (Optional - has no effect unless `cpu_core_count` is also set, **Deprecated** use the `cpu_options` argument instead) If set to 1, hyperthreading is disabled on the launched instance. Defaults to 2 if not set. See [Optimizing CPU Options](#) for more information.

New Multi-language provider docs

Terraform

The Registry now supports multi-language docs powered by CDK for Terraform. [Learn more](#)

ON THIS PAGE

- Example Usage
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- [Argument Reference](#)
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Problemas solucionados?

- Incapacidade de replicabilidade ✅
- Falta de versionamento dos recursos ✅
- Difícil trabalhar colaborativamente ✅
- Falta de documentação escrita — Porquê que criamos cada um dos recursos? ✅
- Processo moroso — minutos! 🟡
- **Difícil "apanhar" modificações aos recursos**

Estado atual

› terraform plan

aws_security_group.enei-security-group: Refreshing **state**... [id=sg-0f3709f1bc4]

aws_instance.virtual-machine: Refreshing **state**... [id=i-0c155532ff05b4f6]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Modificação manual de um recurso

EC2

<

Dashboards

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

EC2 > Instances > i-0c155532ff05b4f6

🔄

📄

Instance summary for i-0c155532ff05b4f6 (enei-presentation-instance) [Info](#)

Updated less than a minute ago

Instance ID

📄 i-0c155532ff05b4f6

IPv6 address

—

Hostname type

IP name: ip-172-31-91-111.ec2.internal

Answer private resource DNS name

—

Auto-assigned IP address

📄 35.171.22.140 [Public IP]

IAM Role

—

IMDSv2

Optional
⚠️ EC2 recommends setting IMDSv2 to required | [Learn more](#)

Operator

—

Public IPv4 address

📄 35.171.22.140 | [open address](#)

Instance state

🟢 Running

Private IP DNS name (IPv4 only)

📄 ip-172-31-91-111.ec2.internal

Instance type

t2.micro

VPC ID

📄 vpc-0527b79bae110e37d

Subnet ID

📄 subnet-0714b7ecdbab05666

Instance ARN

📄 arn:aws:ec2:us-east-1:891377335563:instance/i-0c155532ff05b4f6

Private IPv4 addresses

📄 172.31.91.111

Public IPv4 DNS

📄 ec2-35-171-22-140.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

No recommendations available for this instance.

Auto Scaling Group name

—

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details [Info](#)

AMI ID

📄 ami-0c7217cdde317cfec

AMI name

📄 ubuntu/images/hvm-ssd/ubuntu-jammy-22.04-amd64-server-20231207

Monitoring

disabled

Allowed image

—

Platform details

📄 Linux/UNIX

Termination protection

Disabled

74 / 83

Modificação manual de um recurso

☰ [EC2](#) > [Instances](#) > [i-0c155532ff05b4f6](#) > Manage tags



Manage tags [Info](#)

A tag is a custom label that you assign to an AWS resource. You can use tags to help organize and identify your instances.

Key

Q Name X

Value - *optional*

Q nome-modificado X

Remove

Add new tag

You can add up to 49 more tags.

Cancel

Save

Modificação manual de um recurso - Estado!

```
> terraform plan
```

```
aws_security_group.enei-security-group: Refreshing state... [id=sg-0f3709f1bc466
```

```
aws_instance.virtual-machine: Refreshing state... [id=i-0c1555532ff05b4f6]
```

Terraform will perform the following actions:

```
# aws_instance.virtual-machine will be updated in-place
```

```
~ resource "aws_instance" "virtual-machine" {
```

```
    id                        = "i-0c1555532ff05b4f6"
```

```
    ~ tags                    = {
```

```
        ~ "Name" = "nome-modificado" -> "enei-presentation-instance"
```

```
    }
```

```
    # (39 unchanged attributes hidden)
```

```
}
```

```
Plan: 0 to add, 1 to change, 0 to destroy.
```

Problemas solucionados?

- Incapacidade de replicabilidade ✅
- Falta de versionamento dos recursos ✅
- Difícil trabalhar colaborativamente ✅
- Falta de documentação escrita — Porquê que criamos cada um dos recursos? ✅
- Processo moroso — minutos! 🟡
- **Difícil "apanhar" modificações aos recursos**

Problemas solucionados!!

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- Processo moroso — minutos! 
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Sucesso!



Alternativas

- **OpenTofu** – Fork open-source
 - Primeira versão **stable** lançada em Janeiro de 2024
 - Drop-in replacement de Terraform
- **CloudFormation** + **CDK**:
 - Software proprietário da AWS
 - Possibilidade de escrever código de infraestrutura em linguagens imperativas (Java, Python, TypeScript, ...)
- Pulumi
- Ansible

0 que aprendemos hoje

- O que é a computação em Cloud
- Como utilizar um Cloud provider como a AWS
- Desafios em gerir recursos infraestruturais manualmente
- O que é Infrastructure as Code
- Como utilizar IaC para gerir recursos na AWS
- Vantagens de usar IaC
- Diferentes alternativas disponíveis no mercado

Recursos adicionais

- [Aprender sobre AWS – Recursos gratuitos](#)
- [Documentação oficial do Terraform:](#)
 - [Gestão de estado remoto e colaborativo](#)
 - [Utilização de workspaces para segregar estado](#)
 - [Importar recursos já existentes para o Terraform](#)

Recursos adicionais

Podem encontrar esta apresentação em <https://ruialves.net/enei-2025.pdf>, bem como o código em <https://github.com/xRuiAlves/enei-presentation-2025>.

Obrigado pela vossa atenção!

